

Final Year Project:

Department of Electrical and Electronics Engineering

Signal Denoising and Intelligent Modulation Classification Using Machine Learning

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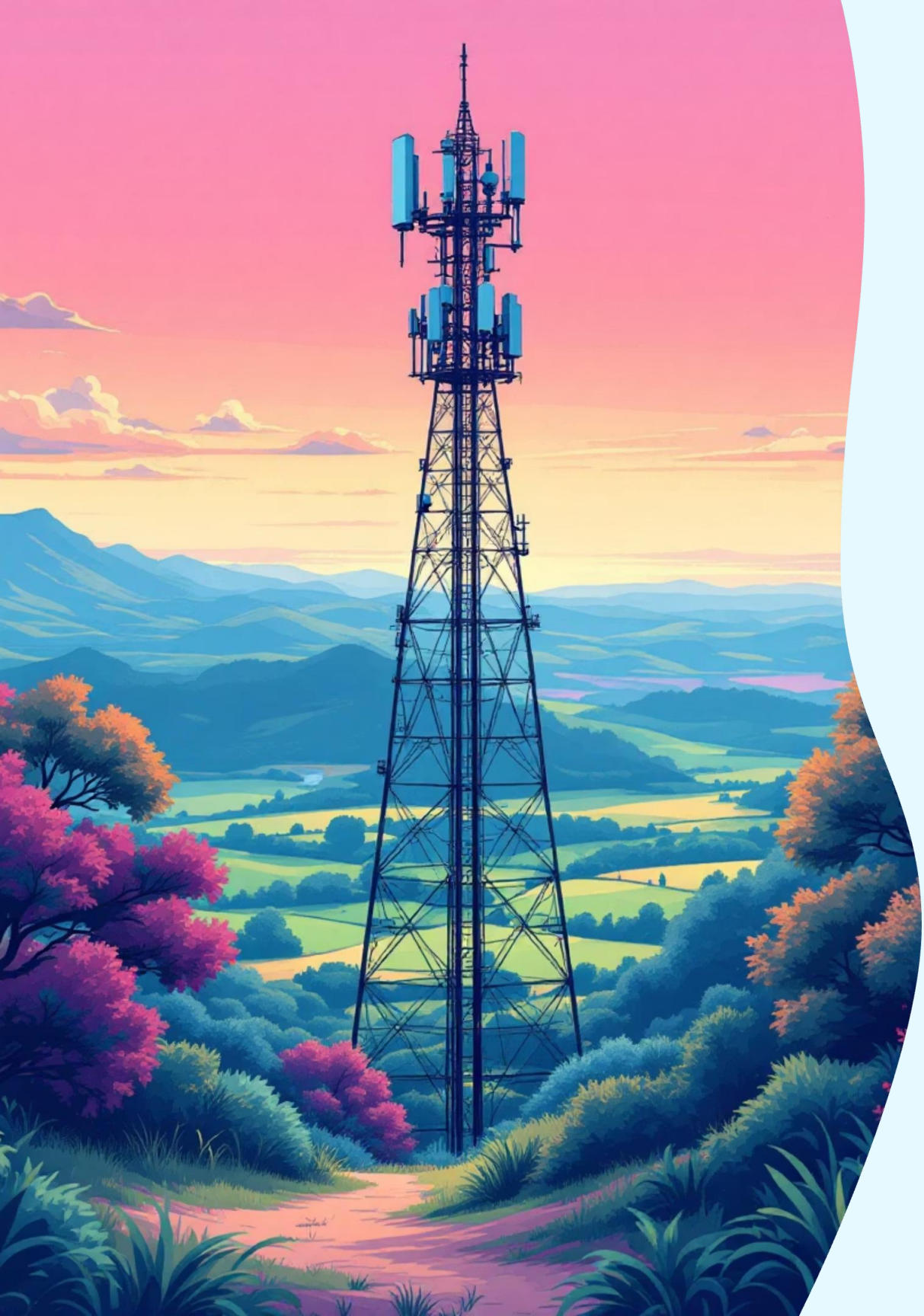
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Proposal Outline

1	Background
2	Problem Statement
3	Objectives
4	Justification
5	Scope of the Project
6	Literature Review

7	Methodology
8	Tools and Equipment
9	Project Timeline
10	Significance
11	Conclusion
12	References

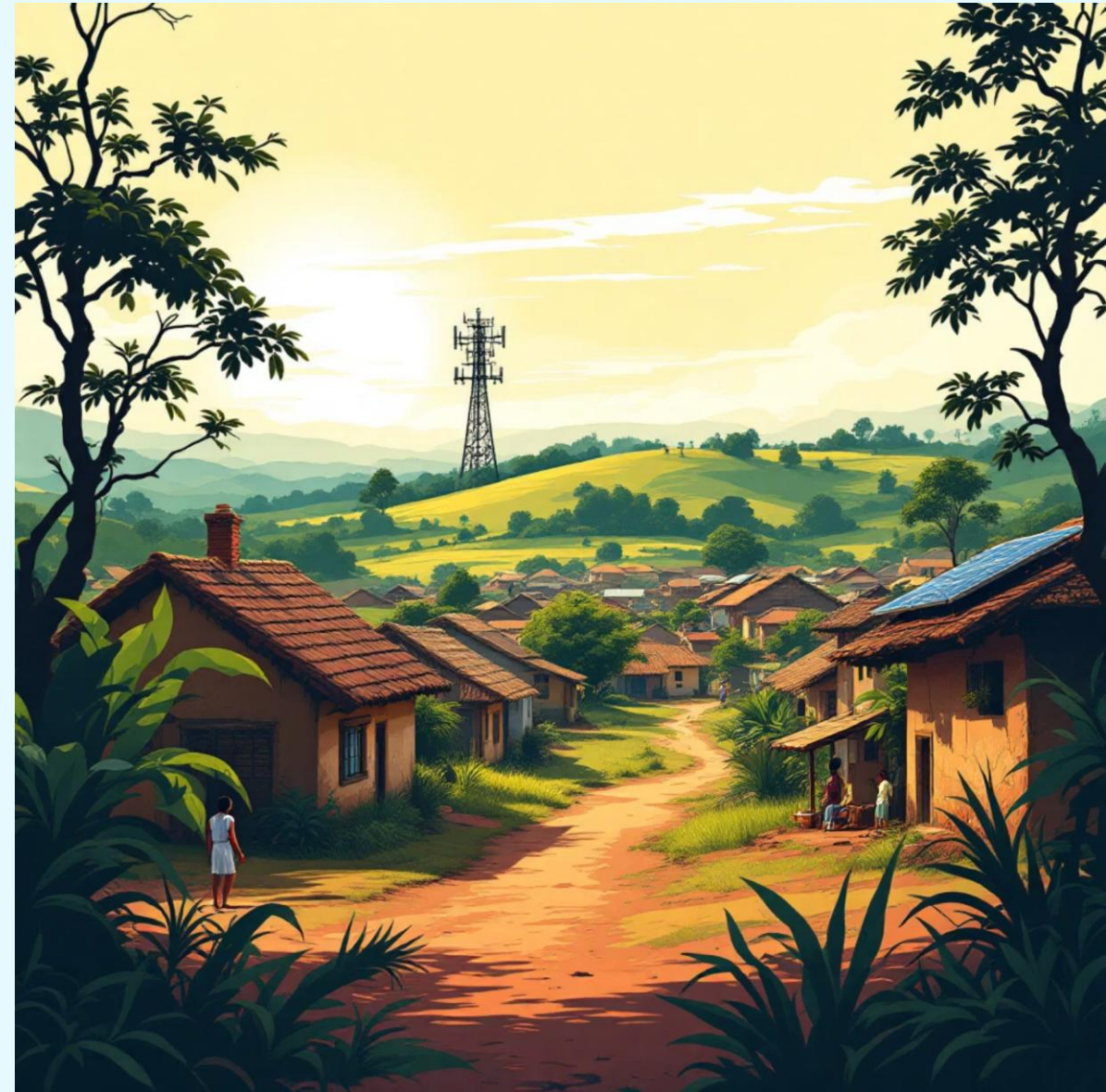


Background

- UCC reports that fixed and mobile subscriptions climbed to **33.2 million** by the end of 2022, **33.1 million** of them mobile, National tele-density is 77 lines per 100 people and mobile data traffic doubled from 217 million GB in 2020 to 421.5 million GB in 2022.
- But network expansion is failing to keep pace with demand; only 31% of the population and 24% of the landmass have 4G coverage, 62% of the population (30 million people) still lacks mobile internet access, and live in low SNR, high-interference environments
- These places require advanced radio technologies to maintain service quality and separate different signals.

Background continuation

- Automatic Modulation Classification (AMC) is crucial for regulators to monitor the spectrum, manage interference, and identify illegal broadcasters that threaten critical services like aviation.
- We propose using a Denoising Autoencoder (DAE) combined with an AMC system to improve signal classification accuracy, especially in low-SNR conditions across Uganda.
- Our Denoising Autoencoder (DAE) takes a marginal signal (e.g., at -90 dBm or even lower) and actively cleans it up removing noise before trying to classify or demodulate it.
- This allows the receiver to maintain a more complex (and faster) modulation scheme than it otherwise could. It improves the reliability and performance at the edge of the cell.

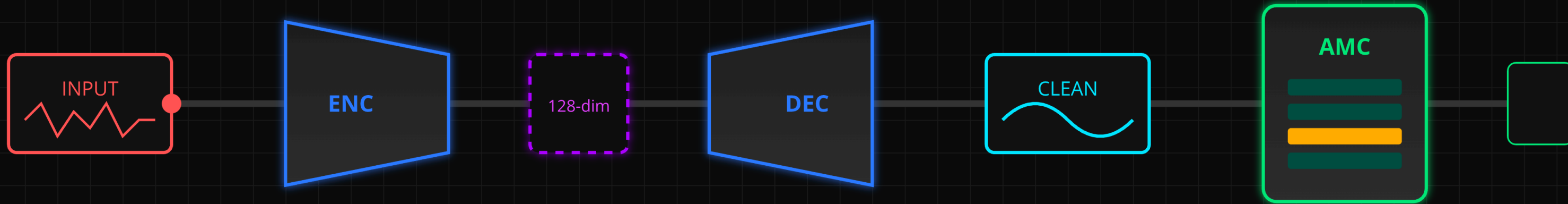


Problem Statement

Uganda's rapid mobile expansion into rural areas is pushing networks to operate at the very edge of their coverage limits (around -90 dBm), creating vast, low Signal-to-Noise Ratio (SNR) environments. This situation is compounded by spectrum congestion and widespread co-channel interference, which the Uganda Communications Commission (UCC) has identified as the cause for over 70% of call failures in audited areas.

Conventional Automatic Modulation Classification (AMC) systems, which are critical for spectrum monitoring, interference mitigation, and identifying illegal transmitters, are not effective in these conditions. Their performance collapses at low SNRs, with even advanced machine learning models failing to achieve the accuracy required for reliable network management and regulatory enforcement.

This creates a critical capability gap: without a robust method to accurately classify modulated signals in high-noise environments, network operators cannot ensure Quality of Service, and regulators cannot effectively police the airwaves to protect critical infrastructure like aviation services. There is an urgent need for a noise-resilient AMC solution that can maintain high accuracy in the challenging, real-world signal conditions found across Uganda.



Main Objective

Design a Hybrid DAE-AMC Pipeline

To design, implement, and empirically evaluate a hybrid **Denoising Autoencoder–Automatic Modulation Classification (DAE-AMC) pipeline** that suppresses interference in noisy I/Q streams and maintains accurate modulation decisions across low SNR channel conditions observed in Ugandan networks.

Specific Objectives

01

Review AMC and DAE literature emphasizing low SNR mitigation strategies relevant to Ugandan channel conditions.

02

Architect and train Conv1D DAE module reconstructing clean I/Q samples from noisy inputs at -90 dBm thresholds. Implement supervised AMC classifier ingesting raw and DAE-cleaned signals, quantifying feature preservation.

03

Benchmark integrated DAE-AMC against standalone AMC using accuracy, F1-score, confusion matrices across various SNR.

Justification and Significance

- ❏ **Meeting UCC Mandates:** The DAE–AMC pipeline addresses UCC's requirement to resolve 95% of faults within 24 hours. These goals are currently unattainable because the Automatic Modulation Classification (AMC) models they rely on are not accurate enough in the low-signal (low-SNR) conditions common across the country, where accuracy can drop to around 72%.
- ❏ **It Aligns with National Policy:** Our project supports Uganda's national development plans (NDPIII/NBP) that call for more spectrum-efficient technologies. It replaces older, "brittle" handcrafted methods with a learnable, AI-based denoiser that is more robust.

Spectrum Surveillance

It will help with the detection of illegal transmitters and interference mitigation for aviation and public-safety channels.

Rural Broadband

It will help maintains reliable service in low-SNR environments, supporting 30M Ugandans lacking mobile internet access.

Critical Services

Ensures stability for mobile money and e-government services dependent on reliable radio links.

Scope



Offline Experimentation

Project conducted entirely on a computer using pre-existing data, ensuring a controlled environment.



Public Datasets

Leveraging well-known I/Q signal datasets like RadioML 2018.01A and MIGOU-MOD for training and testing.



Simulated Noise

Digitally adding controlled AWGN (SNR from -12 dB to +18 dB) to replicate real-world signal degradation.



Model Development

Focus on a Conv1D Denoising Autoencoder (DAE) and a supervised AMC classifier.



Specific Modulations

Training models to recognize BPSK, QPSK, 8PSK, 16QAM, 64QAM, AM, and FM signals.



Platform & Tools

Windows OS, Python, TensorFlow, Keras, Jupyter Notebook, and Google Cloud Platform.



Timeline

A 6-month dedicated period for research, model development, and comprehensive testing.

Literature Review

Chapter Two



Classical Methods

Early Automatic Modulation Classification (AMC) relied on statistical analysis and manually engineered features. While elegant, these methods are brittle and fail completely in the low-signal, high-interference environments common in Uganda.

📄 **Key Limitation:** Brittleness under unknown channels exactly the impairments documented by UCC in congested Ugandan deployments motivates the transition toward data-driven feature learning.

Machine learning is the new standard

Modern research has shifted to using deep learning (DL) models like CNNs, Transformers, and Mixture-of-Experts frameworks. These models can learn features directly from raw signal data and have shown significant accuracy improvements, especially in low Signal-to-Noise Ratio (SNR) conditions down to 0 dB.

Machine Learning in AMC

Deep learning has reshaped AMC by learning discriminative features directly from raw I/Q samples. Recent contributions demonstrate tangible low-SNR gains:

AbdElaziz et al. (2023)

1

Robust CNN with parallel asymmetric kernels achieved 96.5% accuracy at 0 dB and 86.1% at −2 dB across nine modulations.

2

Zhang et al. (2023)

MoE-AMC mixture-of-experts framework yielded ~71.8% averaged accuracy across −20...18 dB—about 10% higher than single-expert models.

Meta-Transformer (2024)

3

Leverages transformer encoders and few-shot learning to adapt rapidly to unseen modulations while maintaining superior accuracy.

4

Rehman et al. (2025)

DLAMC converts I/Q streams into eye diagrams, overcoming the 10–48% accuracy ceiling near −10 dB SNR.

Jagannath et al. (2022)

5

Validated CNN-based multitask AMC on USRP SDR testbed, demonstrating >98% accuracy in live over-the-air experiments.

Denoising Autoencoders for Signal Enhancement

Denoising frontends have emerged as an effective countermeasure when raw I/Q features are overwhelmed by interference.

75%

Zhang et al.

Dual residual DAE with channel attention improved AMC accuracy by up to 75% across $-12 \dots 8$ dB SNR.

77.5%

DenoMAE

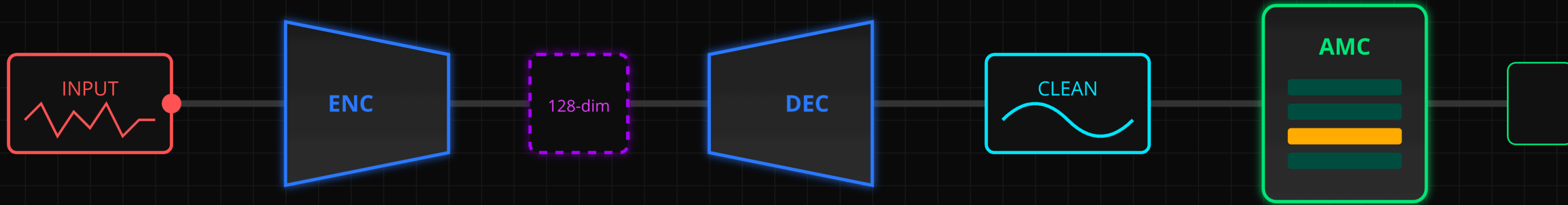
Faysal et al.'s multimodal denoising masked autoencoder sustained 77.5% accuracy at -10 dB—roughly 22% higher without denoising.

70%

Thresholded AE

An and Lee's thresholded autoencoder with SNR predictor delivered $\sim 70\%$ relative accuracy gains on low-SNR samples.

These findings justify the DAE–AMC architecture explored in this project and provide design cues for gating strategies that conserve energy on SDR deployments.



CHAPTER THREE – METHODOLOGY

A comprehensive framework for developing robust Automatic Modulation Classification systems tailored to challenging RF environments through deep learning and denoising techniques.

Introduction to Methods

We adopt an experimental research design anchored in reproducible data processing and quantitative benchmarking across four interconnected pillars:

01

Dataset Curation

Aggregate I/Q datasets from simulated and real captures including RF Signal Data, RadioML 2016.10A, RadioML 2018.01A, MIGOUMOD, and RML22 subsets.

03

Model Training

Train Conv1D denoising autoencoder (unsupervised) and 1D CNN AMC classifier (supervised) both independently and as integrated pipeline.

02

Controlled SNR Scenarios

Generate SNR conditions (−12 to +18 dB) mimicking edge conditions with added impairments: Rician fading, frequency offsets, and timing jitter.

04

Comparative Evaluation

Compare hybrid system against standalone baselines using accuracy curves, F1-scores, and confusion matrices for SDR deployment readiness.

Research Strategy



Dataset Harmonization

Convert datasets to unified tensor format (1024-sample length, normalized amplitude) with stratified train/validation/test splits by modulation and SNR.



Noise Modeling

Inject AWGN, Rician fading, oscillator offsets, and impulsive noise to emulate Ugandan RF environments with comprehensive metadata logging.



DAE Pretraining

Train Conv1D encoder-decoder on noisy/clean pairs using MSE loss until reconstruction PSNR converges, ensuring low-SNR fidelity.



AMC Training

Train baseline on raw inputs, then retrain with DAE outputs to create hybrid pipeline, comparing frozen and fine-tuned configurations.



Cross-Dataset Evaluation

Assess models on held-out SNR bins, unseen modulation families, and OTA-style splits to expose and quantify domain shift effects.



Deployment Prototyping

Integrate trained models into lightweight inference service with GUI for visualization and regulatory demonstrations.

Project Budget Overview

The successful execution of this project requires a structured allocation of resources. The total estimated budget is 800,000 UGX, covering critical areas from software to research.

Item	Cost (UGX)
Software Licenses	200,000
Data Curation	100,000
Virtual Machines	200,000
Cloud Databases	100,000
Internet and Research	50,000
Printing and Documentation	50,000
Miscellaneous	100,000
Total Estimated Cost	800,000

This budget ensures that all necessary technical infrastructure, data processing capabilities, and operational requirements are adequately funded to support the project's objectives.

Project Timeline

Our project will be executed in five distinct phases, ensuring a structured and efficient progression towards our objectives over a 14-week period.

Phase	Description	Weeks	Timeline (Week)
Phase 1: Initiation & Literature Review	Define problem, scope, objectives, and conduct literature review.	1-2	
Phase 2: Data Acquisition & Preprocessing	Gather datasets, implement preprocessing pipelines, and perform feature extraction.	3-5	
Phase 3: Model Development & Training	Select & implement ML algorithms, develop models, train & validate.	6-9	
Phase 4: Testing & Optimization	Evaluate model performance, fine-tune hyperparameters, and optimize.	10-12	
Phase 5: Documentation & Presentation	Compile research findings, methodologies, results, and prepare presentation.	13-14	

This timeline ensures a methodical approach to each stage, culminating in a robust and well-documented solution for intelligent signal processing.

References

- [1]: Uganda Communications Commission, “Telephone Subscriptions Rise to 33.2 Million,” *UCC Communications Blog*, 9 June 2023, <https://uccinfoblog.com/2023/06/09/telephone-subscriptions-rise-to-33-2-million/>.
- [2]: Atomic Energy Council, “Radiofrequency Radiation in Uganda,” 2022, <https://www.atomiccouncil.go.ug/non-ionizing-radiation-radiofrequency/>.
- [6]: Ghana Chamber of Telecommunications, “Mobile Internet Access Still Limited in Africa, Millions Remain Offline,” citing GSMA data, 2024, <https://www.telecomschamber.org/industry-news/mobile-internet-access-still-limited-in-africa-millions-remain-offline/>.
- [7]: Uganda Communications Commission, “UCC cracks down on illegal and non-compliant broadcasters,” 21 October 2024, <https://www.ucc.co.ug/ucc-cracks-down-on-illegal-and-non-compliant-broadcasters/>.
- [8]: Xiaolin Zhang et al., “Dual Residual Denoising Autoencoder with Channel Attention Mechanism for Modulation of Signals,” *Sensors* 23, no. 1023 (2023), <https://pmc.ncbi.nlm.nih.gov/articles/PMC9861137/>.
- [9]: *Low SNR, Interference & Illegal Transmissions in Uganda’s Wireless Networks*, internal research brief, 2025 (synthesizing UCC QoS surveys and enforcement bulletins).
- [10]: *Quantifying the Impact of Low SNR and Interference on Wireless Service Resilience*, internal research brief, 2025 (summarizing NDPIII/NBP mandates and AMC accuracy benchmarks).
- [11]: RF Signal Data, Kaggle, accessed 2025, <https://www.kaggle.com/datasets/suraj520/rf-signal-data>.
- [12]: DeepSig Dataset: RadioML 2018.01A, Kaggle, accessed 2025, <https://www.kaggle.com/datasets/pinxau1000/radioml2018>.
- [13]: Ramiro Utrilla, “MIGOU-MOD: A dataset of modulated radio signals acquired with MIGOU, a low-power IoT experimental platform,” Mendeley Data V1, 2020, <https://data.mendeley.com/datasets/fkwr8mzndr/1>.
- [14]: DeepSig, “RadioML 2016.10A Dataset,” <https://www.deepsig.ai/datasets/>, accessed 2025.
- [15]: O. F. Abd-Elaziz, A. M. El-Ghandour, and F. H. Ismail, “Deep Learning-Based Automatic Modulation Classification Using Robust CNN Architecture for Cognitive Radio Networks,” *Sensors*, vol. 23, no. 23, 2023, Art. 9467, doi:10.3390/s23239467.
- [16]: J. Gao, Z. Zhang, and Y. Zhang, “MoE-AMC: Enhancing Automatic Modulation Classification Performance Using Mixture-of-Experts,” *arXiv preprint*, 2023, <https://arxiv.org/abs/2312.02298>.

Thanx for Listening